DESIGN DOCUMENT

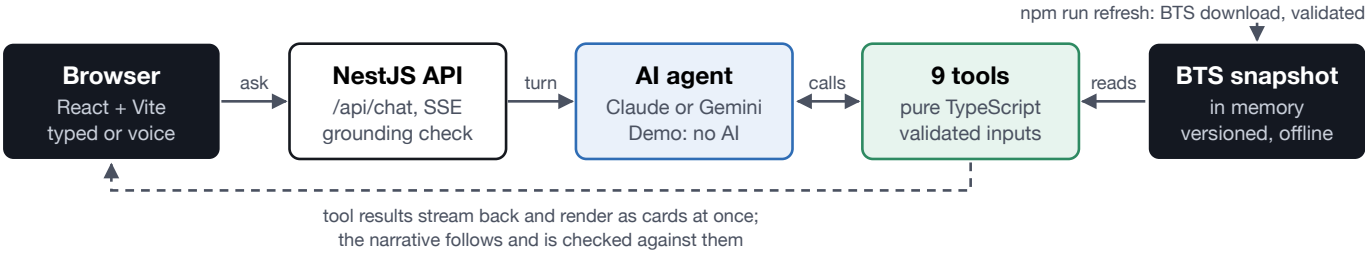
Airport Investment Intelligence Agent

A chat workspace for airport investors: an AI agent picks the analysis, deterministic TypeScript tools compute every figure from official U.S. DOT / BTS data.

Date	3 November 2026	**Data snapshot**	2026-11-03
Stack	React + Vite, NestJS, TypeScript	**AI**	Claude, Gemini, Demo (no AI)

1 Overview and architecture

Analysts ask in plain language: which U.S. airports rank highest for investment nationally or in a region such as New England, Anchorage's long-haul share, LAX vs SNA delays, unmet demand at SFO, or a profile of any airport. **The language model never produces a number.** It chooses tools and writes a short narrative; every figure, rank, score, source and date comes from deterministic code over a versioned snapshot of official BTS data (T-100 traffic Jan 2019 – Jul 2026, route-level T-100, delay causes).



2 Scoring methodology and eligibility

```
// P = departing passengers, summed over 12 months (BTS r495); ratios from summed counts
Growth %      = 100 × (P2025 − P2024) / P2024
Load factor % = 100 × P2025 / Seats2025
GrowthScore   = clamp(50 + 2.5 × Growth %, 0, 100)            // 0% → 50, +20% → 100, −20% → 0
LoadScore     = clamp((Load factor % − 60) / 30 × 100, 0, 100)  // 60% → 0, 75% → 50, 90% → 100
OverallScore  = 0.60 × GrowthScore + 0.40 × LoadScore
```

Eligibility: all must hold

≥ 500,000 departing passengers in 2025	`BELOW_THRESHOLD`
12 reported months in 2024 and 2025	`INCOMPLETE_2025`, `_2024`
2024 passengers > 0	`NO_2024_BASELINE`
2025 seats > 0, passengers ≤ seats	`LF_UNAVAILABLE`, `INVALID_LOAD_FACTOR`

Rules

- All U.S. airports incl. territories; regions keep national ranks.
- Missing months are never filled; delays and 2026 data are not scored.
- Excluded airports list every failing reason.

Result (pinned by golden tests): 144 airports ranked; PGD first (90.7), HVN first in New England (#7 nationally), OAK last (25.5). Growth spans −14.7% to +25.3%, load factor 65.8% to 86.0%.

3 Where AI is used, and where it is not

AI (Claude or Gemini, chosen in the chat)

- Understands questions and follow-ups.
- Picks tools and arguments; asks when the airport, region, metric or period is missing.
- Writes a 2 to 6 sentence narrative.

Deterministic code

- Every figure, rank, score, eligibility decision, source and date.
- Cards, tables and charts, rendered from tool payloads only.
- **Grounding check:** each number in the narrative must match a tool result, otherwise the UI shows a warning.
- Demo mode: seven fixed analyses, no AI at all.

4 Key tradeoffs

Decision	Instead of	Why, and the cost
Model selects tools, code computes	Model computes or summarises raw data	Auditable, reproducible numbers. Cost: questions outside the 9 tools are declined rather than improvised.
Versioned snapshot in memory	Live BTS queries per request	Identical results offline, sub-millisecond tools, no fragile TranStats forms at request time. Cost: data is as fresh as the last refresh; a freshness check flags newer BTS data but never changes numbers.
Simple, transparent score: two measured signals, 60/40, 500k threshold, no imputation	Richer model with delays, capacity or forecasts	Both inputs are measured by BTS; growth is the primary signal. The threshold keeps small bases from dominating (BID +61.8% on 19,855 passengers). Cost: financials and terminal capacity are not captured.
Claude, Gemini and Demo, chosen per conversation	One AI provider	Two interchangeable agents over the same tools, and a Demo mode that runs without any key. Cost: two agent implementations; a conversation stays with one provider because their histories cannot be mixed.

5 Assumptions and limitations

Terminal expansion is ranked by demand pressure, not terminal capacity. BTS publishes no terminal, gate, check-in or TSA throughput capacity. Load factor is the share of aircraft seats filled, not terminal utilisation. A high score means strong passenger growth and full flights; it does not show that a terminal is out of space, and a low score does not show spare space.

Unmet demand at SFO cannot be quantified from BTS data. BTS records realised traffic, the passengers who flew. Denied or suppressed demand, travellers priced out, slot and runway limits and demand spilling to OAK or SJC are not observed. The app shows proxies instead: 2025 passengers –5.0% vs 2019 and +4.7% vs 2024, passenger-service departures –11.1% vs 2019, seats per departure, and peak-month load factors (June 2024: 91.2%). High load factors with fewer flights are consistent with capacity pressure but are not proof of unmet demand.

- **Passengers** are departing (enplaned) passengers from T-100, not airport footfall. **Passenger flights** are scheduled or charter service classes with passengers > 0; cargo-only flights are excluded. **Long-haul** is ≥ 3,000 statute miles (BTS has no definition).
- **Delays** cover domestic arrivals of U.S. reporting carriers only, so they are a diagnostic of punctuality, not a direct measure of congestion.
- **2026 data is partial** (traffic to July, delays to August) and is compared only with the same months of 2025. Results are tied to the recorded BTS vintage; BTS may revise past months.

Details in the repository: `README.md` (setup, sample questions), `docs/SCORING.md`.